MEDICAL EMERGENCIES MODULE

What to remember

Online Module Quick Reference

Each lesson of the online module is distilled to its four or five essential ideas. This document does not replace the manual, the video capsules or in-class practice.

For every medical emergency, the approach is the same: complete PAS, SAMPLE history first since the patient is ill rather than injured, serial vital signs, then the question that guides everything: what type of shock can this emergency produce, and is evacuation needed?

MODULE CONTENTS: ELEVEN LESSONS

00 Introduction to the chapter

Ten lessons, grouped into four families, to recognize what is happening inside

> **A medical emergency has no visible mechanism of injury.** It is the history, the medical background and the trend in vital signs that tell you what is happening, and which type of shock is threatening.

1. **Four families to organize your reasoning.** Vital systems (heart, brain, lungs), neurological and metabolic (epilepsy, diabetes), digestive system (acute abdomen, gastroenteritis), reactions and substances (allergies, poisons, medications).
2. **Ill patient: medical history first.** SAMPLE before the physical exam. The symptoms reported by the patient are your subjective data; what you measure and observe is your objective data. Record both, and any discrepancies.
3. **Always ask yourself: which shock?** Each emergency in this module can progress to a specific type of shock (cardiogenic, hypovolemic, distributive, respiratory). Recognizing it early guides positioning, monitoring and the urgency of evacuation.
4. **The Field Manual contains a SAMPLE for each emergency.** Use it as a questioning guide during the assessment, not just as a reference afterward.
5. **Medications: know that they exist, what they are for, and your limits.** The first aider helps the patient take their own prescribed medication. The first aider does not prescribe. Exceptions are covered in class, case by case, in the scenarios.

LEARNER'S MANUAL	FIELD MANUAL	IN CLASS
Section 11, Medical Emergencies · Online encyclopedia	**Pages 192 to 216**	**Assessment and management of each emergency, nine scenarios from the participant workbook**

01 Cardiovascular emergencies

Recognizing chest pain and the signs of a heart problem

> **Angina, heart attack, cardiac arrest: one continuum.** Angina is an insufficient supply of oxygenated blood to the heart; a myocardial infarction is death of heart muscle; cardiac arrest is the heart's inability to contract. The associated shock is cardiogenic.

1 Repos et position
Cesser tout effort,
semi-assise, DEA prêt

→ **2 AAS à croquer**
325 mg ou 2 × 81 mg,
le plus tôt possible

→ **3 Nitro : la sienne seulement**
Angineux connu, prescrite,
jamais si signes de choc

Figure 1 — L'ordre compte : l'AAS améliore la survie, la nitroglycérine soulage seulement.

1. **The signs that should alert you.** Heaviness, tightness or pressure in the chest lasting 2 to 15 minutes, radiating to the left arm, back, jaw or both arms, with nausea, profuse sweating, weakness, shortness of breath and often denial. Women, older adults and people with diabetes sometimes present atypically.
2. **Angina or heart attack: rest decides.** Angina is relieved by rest and by the prescribed medication. Pain that persists despite rest, returns quickly or comes with signs of shock suggests a heart attack: activate emergency services or evacuation without delay.
3. **ASA first, as early as possible.** Aspirin is the only medication shown to improve survival in a heart attack. Help the patient chew 325 mg (or two 81 mg tablets), unless there is a known allergy or contraindication. In the backcountry, with long delays, recommending it is reasonable: very low risk, high benefit.
4. **Nitroglycerin: only for the patient who has their own.** It does not improve survival and can cause significant hypotension. Reserved for people with known angina who have their prescription and know how to use it: never in the presence of signs of shock, nor within 48 hours of a vasodilator (Viagra, Cialis, Levitra).
5. **Strict rest, position, monitoring, AED ready.** Stop all activity, semi-sitting or position of comfort, oxygen if available, serial vital signs. If the patient loses consciousness and stops breathing: CPR and AED. Hypertensive emergency: blood pressure of 180/120 or higher with symptoms.

LEARNER'S MANUAL
Section 11.1, Cardiovascular Disease · Online encyclopedia

FIELD MANUAL
Pages 192 and 193

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02 Cerebrovascular emergencies

Quickly spotting the signs of a stroke and acting on time

A stroke is an interruption of blood flow to part of the brain, by blockage (ischemic) or by rupture of a vessel (hemorrhagic). A TIA produces the same signs, but they resolve in less than 24 hours: it is a warning, not a false alarm.

1. **Anything sudden is suspect.** Sudden weakness or numbness of the face or a limb, slurred speech, vision disturbance, loss of balance, sudden and unusual headache, dizziness, altered level of consciousness.
2. **FAST: screening in three steps.** Face: ask for a smile, look for asymmetry. Arms: ask the patient to raise both arms, one drifts or does not rise. Speech: have them repeat a simple sentence. Time: a single sign is enough to act.
3. **Every minute counts, really.** Hospital treatment is most effective in the first hours after symptom onset. Note the exact time of onset (or the time last seen normal): it is the most important information to pass on.
4. **No aspirin.** Unlike a heart attack, you cannot distinguish an ischemic stroke from a hemorrhagic stroke in the field. ASA would worsen a brain hemorrhage.
5. **Position, airway, evacuation.** Position of comfort, recovery position if altered level of consciousness or vomiting, nothing by mouth (swallowing difficulty is common), oxygen if available, be ready for CPR. Risk factors to know: hypertension, diabetes, smoking, previous stroke or TIA.

LEARNER'S MANUAL	FIELD MANUAL	IN CLASS
Section 11.2, Cerebrovascular Disease · Online encyclopedia	**Pages 194 and 195**	**Stroke and TIA scenario, Cincinnati criteria**

03 Respiratory emergencies

Distinguishing the causes of respiratory distress and supporting the patient

Respiratory distress is a direct threat to oxygen, the first of the seven essential elements. The associated shock is respiratory or obstructive. Asthma is its most frequent medical cause in the backcountry.

1. **Recognizing ineffective breathing.** Fast or very slow rate, visible effort, accessory muscles, pursed lips, incomplete sentences, wheezing on exhalation, cyanosis, agitation then drowsiness. A patient who calms down without improving is becoming exhausted: this is a sign of severity.
2. **Asthma: inflammation, mucus, spasm.** The bronchi become inflamed, secrete mucus and constrict. Frequent triggers: exertion, cold, allergens, respiratory infection, smoke, stress. Exhalation becomes long and wheezy, the chest expands.
3. **The patient's medication, right away.** Help them take their rescue bronchodilator (blue inhaler, salbutamol type) as prescribed; it works within minutes. Inhaled corticosteroids are a maintenance treatment, not a rescue medication.
4. **Position and calm.** Sitting, leaning forward, arms supported. Reassure, coach slow breathing, move away from the trigger, warm the inhaled air in the cold. Serial vital signs, especially respiratory rate and ability to speak.
5. **Evacuate if the inhaler is not enough.** No improvement after medication, repeated need for doses, cyanosis, inability to speak, altered level of consciousness: urgent evacuation. Also consider the other causes of dyspnea: anaphylaxis, pneumonia, pneumothorax, embolism, anxious hyperventilation.

LEARNER'S MANUAL	FIELD MANUAL	IN CLASS
Section 11.6, Non-Traumatic Respiratory Emergencies · Online encyclopedia	Pages 204 and 205	Asthma scenario, straw activity

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04 Epilepsy and seizures

Protecting the patient during the seizure, knowing what to observe

You can have a seizure without having epilepsy. Epilepsy is a chronic condition; an epileptic seizure is an abnormal electrical discharge; convulsions are its muscle contractions. Fever, hypoglycemia, head injury, intoxication and lack of sleep can all trigger a seizure.

1. **Partial or generalized.** A partial seizure affects one specific function (motor, sensory, behaviour) with or without altered consciousness. A generalized tonic-clonic seizure goes through a rigid phase followed by jerking, usually lasts less than 2 minutes, sometimes with incontinence and a brief pause in breathing.
2. **During the seizure: protect, time it, force nothing.** Move dangerous objects away, protect the head, do not restrain the person, put nothing in the mouth. Note the start time and the duration: it is your most useful piece of data.
3. **After the seizure: the postictal phase.** Confusion, fatigue, disorientation lasting from a few minutes to a few hours. Recovery position, open airway (snoring is common), monitoring of breathing and consciousness, rewarming, gradual return to normal.
4. **The signals for urgent evacuation.** Seizure lasting more than 5 minutes or repeated seizures without regaining consciousness in between (status epilepticus), first seizure ever, seizure after a head injury, injury during the seizure, pregnancy, diabetes, or failure to return to normal.
5. **Before departure: know who is at risk.** Type of seizure, triggers, frequency, medication and side effects, personal plan. A febrile seizure in a child (6 months to 6 years) is managed with ABC monitoring and fever control.

LEARNER'S MANUAL	FIELD MANUAL	IN CLASS
Section 11.3, Seizures · Online encyclopedia	**Pages 196 and 197**	**Epilepsy and febrile seizure scenario**

05 Diabetes

Recognizing blood sugar swings and their misleading signs

Glucose needs the insulin key to enter the cell. Type 1: the pancreas no longer produces insulin. Type 2: it does not produce enough or the insulin works poorly. The two emergencies are hypoglycemia (fast, dangerous) and hyperglycemia (slow, serious).

1. **Hypoglycemia looks like anything but diabetes.** Rapid onset: hunger, shakiness, sweating, pale, cool and clammy skin, rapid pulse, headache, confusion, agitation, slurred speech, strange behaviour, seizures, loss of consciousness. It mimics drunkenness, stroke or head injury.
2. **Hyperglycemia sets in over hours or days.** Intense thirst, frequent urination, weakness, nausea, vomiting, fruity (acetone) breath, warm dry skin, deep rapid breathing, dehydration, then altered level of consciousness. Treatment is medical: hydration and evacuation.
3. **When in doubt, give sugar.** A simple sugar improves hypoglycemia within minutes and makes almost no difference to hyperglycemia. Conscious patient: simple sugar, then a full meal if they improve. Unconscious patient: recovery position, glucose gel rubbed inside the cheek, evacuation.
4. **The glucometer confirms; it does not replace clinical judgment.** If one is available and the patient or a companion knows how to use it, measure before and after. Record the values and the time on the PAS form.
5. **Nasal glucagon (Baqsimi) is part of the patient's plan, not your kit.** A person with type 1 diabetes on an expedition should have it and should have shown you their management plan before departure. That is what the pre-trip meeting and the medical form are for.

LEARNER'S MANUAL
Section 11.4, Diabetes · Online encyclopedia

FIELD MANUAL
Pages 198 and 199

IN CLASS
Hypoglycemia scenario, glucometer awareness

06 Acute abdominal emergencies

Assessing abdominal pain and recognizing the signs of severity

> **An acute abdomen is severe abdominal pain of sudden onset.** Infection (appendicitis, peritonitis), obstruction, perforation or internal bleeding: in the field, you will not be able to tell which. Your job is to recognize severity and decide on evacuation.

1. **Characterize the pain with SAMPLE and OPQRST.** Onset, location, migration, type (cramping, burning, stabbing), intensity, what makes it worse or better, associated symptoms. Diffuse pain that later localizes is a classic picture of appendicitis.
2. **The red flags that require evacuation.** Blood in vomit, stool (black or red) or urine; fever with vomiting; rigid or distended abdomen, or guarding on palpation; pain that prevents moving or walking; signs of shock; absent bowel sounds; hernia.
3. **All cases of acute abdomen are evacuated.** When in doubt, initiate evacuation. The risk of complications is high and the delay in the backcountry is measured in hours. Abdominal pain lasting more than a few hours without an obvious diagnosis is not something to wait out.
4. **Position of comfort, nothing by mouth, no walking.** Most often semi-sitting or on the side with knees bent, to relax the abdominal wall. Nothing solid by mouth; small sips of clear fluids if the evacuation is prolonged. Litter transport.
5. **Monitor, treat for shock, document.** Serial vital signs, repeated palpation by the same person, record any change. Shock may be hypovolemic (internal bleeding, vomiting) or septic (infection).

LEARNER'S MANUAL	FIELD MANUAL	IN CLASS
Section 11.5, Non-Traumatic Abdominal Emergencies · Online encyclopedia	**Pages 202 and 203**	**Acute abdominal condition scenario**

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07 Gastroenteritis

Managing dehydration and limiting transmission within the group

An infection of the stomach and intestines by viruses, bacteria or parasites, most often transmitted through fecal contamination of water, food and hands. On an expedition, its real danger is twofold: dehydration and spread through the group.

1. **The usual picture.** Nausea, vomiting, diarrhea, cramps, sometimes fever and body aches. The associated shock is hypovolemic: loss of fluids and electrolytes. A child can lose 10 to 20% of circulating volume in under two hours.
2. **Rehydration is the treatment.** Small, frequent amounts rather than large volumes. Oral rehydration solution preferably, otherwise water with a little sugar and salt, or broth. Resume light eating as soon as tolerated. Watch urine colour and output.
3. **The red flags.** Inability to keep fluids down, blood in the stool, high fever, localized abdominal pain or rigidity, signs of severe dehydration (rapid pulse, dizziness on standing, little urine, confusion), symptoms lasting more than a few days.
4. **Protect the rest of the group.** Rigorous hand washing, food prepared by someone who is not ill, individual dishes and water bottles, water disinfection, waste management away from water sources, relative isolation of the ill person.
5. **Loperamide and antibiotics: introduction only.** Loperamide (Imodium) slows transit and can help while on the move, but it is not recommended in the presence of fever or blood in the stool. Antibiotics are a medical decision or an expedition protocol matter, not the first aider's alone.

LEARNER'S MANUAL	FIELD MANUAL	IN CLASS
Section 11.5, Non-Traumatic Abdominal Emergencies · Online encyclopedia	**Pages 200 and 201**	**Gastroenteritis scenario**

08 Allergic reactions and anaphylaxis

Recognizing anaphylaxis and acting without delay

Histamine explains everything. Released locally, it causes redness, itching and swelling. Released massively, it dilates vessels, lets plasma leak out and constricts the bronchi: that is anaphylaxis and its distributive shock.

1. **Local reactions: bothersome, rarely dangerous.** Rhinitis, localized hives, contact dermatitis (poison ivy, sumac), local reaction to a sting. Management: remove the allergen, wash, local cold, antihistamine. Watch the progression: a local reaction can precede a generalized one.
2. **Recognizing anaphylaxis.** Severe respiratory distress or circulatory failure, OR at least two of the following four pictures: hives or angioedema (lips, tongue, eyelids), difficulty breathing, signs of shock (weakness, dizziness), digestive symptoms (nausea, vomiting, cramps, diarrhea).
3. **Epinephrine, immediately, intramuscularly.** Auto-injector (EpiPen) into the outer thigh, through clothing if necessary. When life is at risk, there is no contraindication. A second dose may be needed after 5 to 15 minutes if signs persist or return.
4. **Only afterward: antihistamine, position, shock.** Fast-acting antihistamine (diphenhydramine) if the patient can swallow, never a slow-release formulation. Lying with legs raised if in shock, sitting if breathing is difficult. Treat for shock, keep warm.
5. **Always evacuate, even if everything improves.** Signs can return several hours later (biphasic reaction). Close monitoring for 24 hours and evacuation to a medical facility. Know before departure who has an allergy and where their auto-injector is.

LEARNER'S MANUAL	FIELD MANUAL	IN CLASS
Section 11.6, Non-Traumatic Respiratory Emergencies (anaphylaxis) · Online encyclopedia	**Pages 206 to 209**	**Anaphylaxis scenario, how to use the auto-injector**

09 Poisons and toxins

Identifying the exposure, limiting absorption, guiding management

> **Three routes of entry, three management logics.** Ingestion: prevent absorption. Inhalation: get out of the toxic atmosphere. Skin contact: remove and rinse. In every case, the poison control centre is your ally at a distance.

LES 16 RUBRIQUES

1	**2**	3	**4**
5	**6**	7	**8**
9	10	11	12
13	14	15	16

En couleur : les cinq rubriques utiles au secouriste.

- **1 Identification.** Nom exact du produit et coordonnées du fournisseur : c'est ce que le centre antipoison demande en premier.
- **2 Identification des dangers.** Pictogrammes, mention d'avertissement et conseils de prudence : ce à quoi vous avez affaire, en un regard.
- **4 Premiers soins.** Mesures d'urgence par voie d'exposition : inhalation, contact cutané ou oculaire, ingestion. La rubrique à lire en premier.
- **6 Déversement accidentel.** Précautions individuelles avant d'approcher, et méthode de nettoyage.
- **8 Contrôle de l'exposition.** Équipement de protection individuelle requis : gants, écran facial, protection respiratoire.

Figure 2 — Fiche de données de sécurité (FDS, SIMDUT et SGH) : ordre normalisé des 16 rubriques. Les autres (3, 5, 7 et 9 à 16) relèvent de la prévention, du transport et de la réglementation, pas de l'intervention.

1. **Ingestion: identify, call, do not induce vomiting.** What, how much, when, in whom. Keep the container or a sample. Do not induce vomiting. Contact the poison control centre as soon as communication is possible and evacuate.
2. **Activated charcoal is not a prehospital tool.** It binds the poison in the digestive tract before absorption, but administering it belongs in a hospital setting: it requires a medical assessment, a protected airway, and doses (adult 50 to 100 g) that do not match the formats sold in pharmacies. It is in any case useless or contraindicated with strong acids and bases, metals, alcohol and hydrocarbons, and in a drowsy patient. In the field: identify the product, reach the poison control centre, evacuate.
3. **Carbon monoxide: the invisible killer in shelters.** Colourless, odourless, produced by any combustion in an enclosed space (stove inside the tent, snow shelter, poorly ventilated cabin). Headache, nausea, dizziness, weakness, confusion, often in several people at once. Get everyone out into fresh air, oxygen if available, evacuate.
4. **Skin contact: remove, rinse for a long time.** Remove contaminated clothing while wearing gloves, brush off dry powders before wetting, rinse thoroughly with water. Protect yourself first.
5. **Prevent: know what the group is carrying and what the environment contains.** Medications, fuels, water disinfection products; local plants, mushrooms, insects, snakes. The emergency plan should include the poison control centre number for the region visited.

LEARNER'S MANUAL	FIELD MANUAL	IN CLASS
Section 11.7, Poisons and Toxins · Online encyclopedia	**Pages 210 and 211**	**Poison and toxin scenario, carbon monoxide scenario**

10 Medications

Knowing your scope of action and your limits as a first aider

> **The first aider is not a prescriber.** This lesson teaches you what exists, what it is for, and within what framework you can help a patient take a medication. Any medication use should be supervised by a physician or governed by a protocol.

1. **Three distinct situations.** Helping the patient take their own prescribed medication (nitro, inhaler, auto-injector); recommending an over-the-counter medication that the patient decides to take (ASA, antihistamine, analgesic); administering an emergency medication under protocol or medical direction. Your role changes at each level.
2. **The medications encountered in this module.** ASA for suspected heart attack, nitroglycerin (their own, known angina), rescue bronchodilator, epinephrine, antihistamines, glucose gel, nasal glucagon, activated charcoal, rehydration solution, loperamide. Know their indication, their form and their limits.
3. **The five checks before helping.** Right person, right medication, right dose, right route, right time. Check for allergies, known contraindications and what has already been taken (the M and the A of SAMPLE).
4. **Document, always.** Name, dose, time, route, observed effect and side effects, on the PAS form. This is the information medical services will ask you for first.
5. **Decision-making in scenarios.** In class, each scenario will have you decide whether a medication is relevant, which one, and given by whom. The general rule remains: when in doubt, teleconsultation or evacuation takes precedence over improvisation.

LEARNER'S MANUAL	FIELD MANUAL
N/A	Page 216